MATH 101 Calculus I Name: _____

Problem M1. Integration by parts

Evaluate the indefinite integral $\int x\,e^{2x}\,dx$ using integration by parts.

By parts: $\int u\,dv = uv - \int v\,du$

Let $u = x$, $dv = e^{2x}\,dx$

$du = dx$, $v = 2e^{2x}$

$$\int x\,e^{2x}\,dx = x \cdot 2e^{2x} - \int 2e^{2x}\,dx$$

$$= 2xe^{2x} - 2 \cdot \frac{1}{2}e^{2x} + C$$

$$= 2xe^{2x} - e^{2x} + C$$

Answer: $(2x-1)e^{2x} + C$

Problem M2. Tangent line

Find the equation of the tangent line to the curve $y = \sin(x^2)$ at the point where $x = \sqrt{\pi}$.

$y = \sin(x^2)$, point at $x = \sqrt{\pi}$

$y(\sqrt{\pi}) = \sin(\pi) = 0$, so the point is $(\sqrt{\pi}, 0)$

$\frac{dy}{dx} = \cos(x^2)$

slope at $x = \sqrt{\pi}$: $m = \cos(\pi) = -1$

tangent: $y - 0 = -1(x - \sqrt{\pi})$

Answer: $y = -x + \sqrt{\pi}$

Problem M3. Area between curves

Find the area of the region enclosed by $y = x$ and $y = x^2$ for $0 \le x \le 1$. Sketch the region.

Curves meet where $x^2 = x$: $x = 0$ and $x = 1$

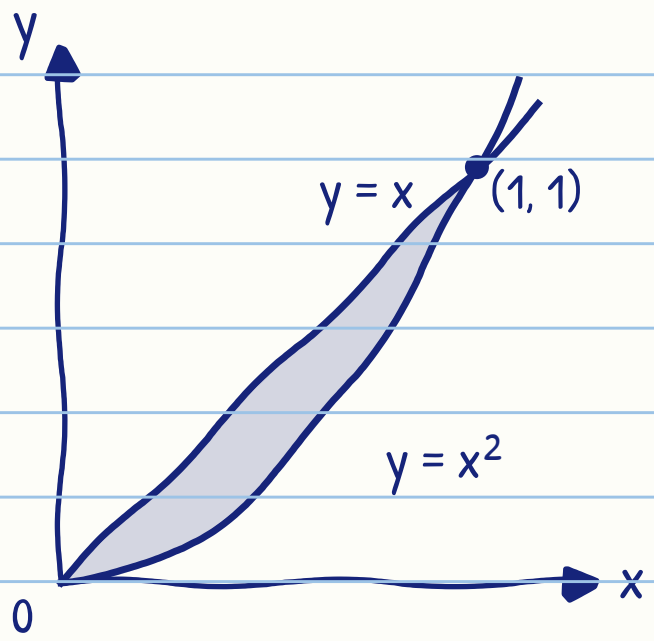
Area $A = \int_0^1 (x^2 - x)\,dx$

$$= \left[\frac{x^3}{3} - \frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{3} - \frac{1}{2} - 0$$

$$= -\frac{1}{6}$$

Answer: area $= -1/6$



Problem M4. Eigenvalues of a 2x2 matrix

Find the eigenvalues of the 2×2 matrix A with rows $(2,\ 1)$ and $(1,\ 2)$.

Eigenvalues: $\det(A - \lambda I) = 0$

$A - \lambda I$ has rows $(2-\lambda,\ 1)$ and $(1,\ 2-\lambda)$

$\det = (2-\lambda)(2-\lambda) + (1)(1)$

$= \lambda^2 - 4\lambda + 4 + 1 = \lambda^2 - 4\lambda + 5$

$$\lambda = \frac{4 \pm \sqrt{16 - 20}}{2} = \frac{4 \pm \sqrt{-4}}{2}$$

$$= \frac{4 \pm 2i}{2}$$

Answer: $\lambda = 2 \pm i$ (complex eigenvalues)

Problem M5. Separable initial value problem

Solve the initial value problem $y' = 2y$, $y(0) = 3$, and sketch the solution for $0 \le x \le 1$.

$$\frac{dy}{dx} = 2y$$

separate: $\frac{dy}{y} = 2\,dx$

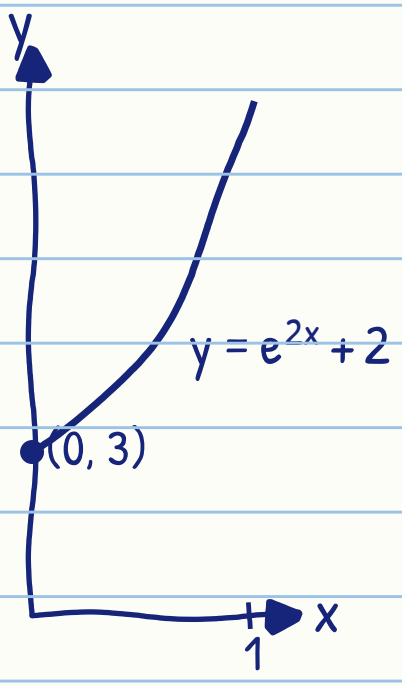
$$\int \frac{dy}{y} = \int 2\,dx$$

$$\ln y = 2x + C$$

$$y = e^{2x} + C$$

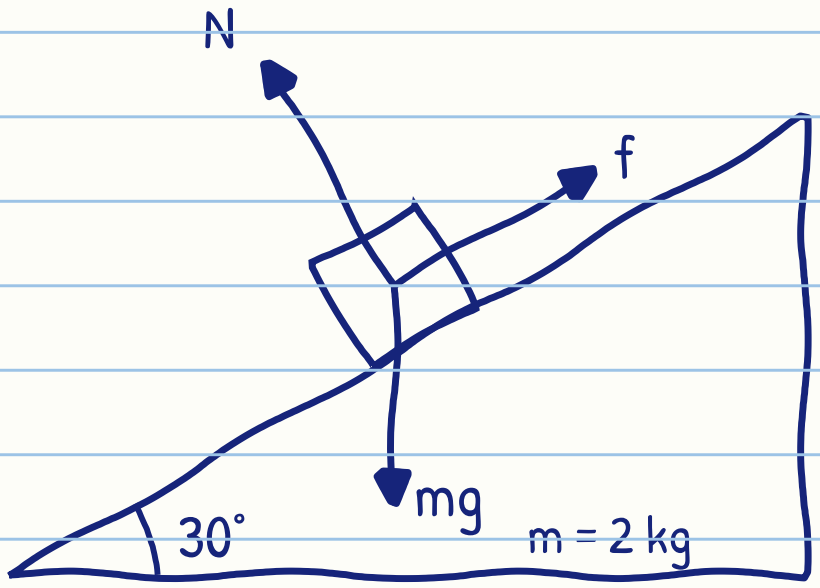
$y(0) = 3$: $e^0 + C = 3$, so $C = 2$

Answer: $y = e^{2x} + 2$



Problem P1. Block on a rough incline

A 2 kg block slides down a rough incline at $\theta = 30°$ to the horizontal. The coefficient of kinetic friction is $\mu_k = 0.2$. Draw a free-body diagram and find the acceleration of the block. Take $g = 9.8$ m/s^2.



Forces: weight mg, normal N, friction f up the slope

Along the slope: $ma = mg\sin\theta - f$

$N = mg = 2(9.8) = 19.6$ N

$f = \mu_k N = 0.2(19.6) = 3.92$ N

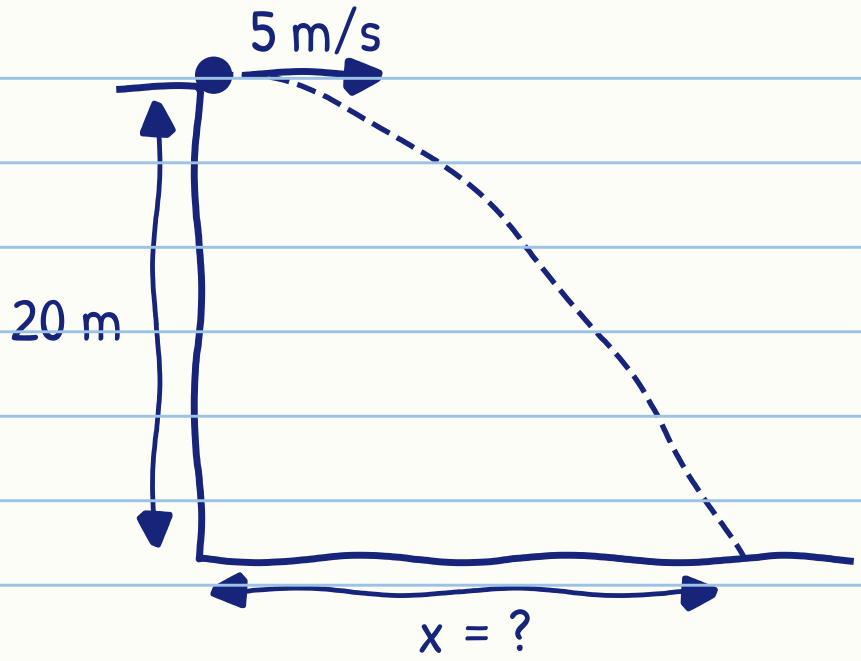
$mg\sin 30° = 19.6(0.5) = 9.8$ N

$2a = 9.8 - 3.92 = 5.88$

Answer: $a = 2.94$ m/s^2 down the slope

Problem P2. Ball thrown horizontally from a cliff

A ball is thrown horizontally at 5 m/s from the top of a 20 m high cliff. How long does it take to reach the ground, and how far from the base of the cliff does it land? Ignore air resistance; take $g = 9.8$ m/s^2.



Vertical, down = +: $y = v_0 t + \frac{1}{2} g t^2$

$20 = 5t + 4.9t^2$

$4.9t^2 + 5t - 20 = 0$

$$t = \frac{-5 + \sqrt{25 + 392}}{9.8} = \frac{-5 + 20.42}{9.8}$$

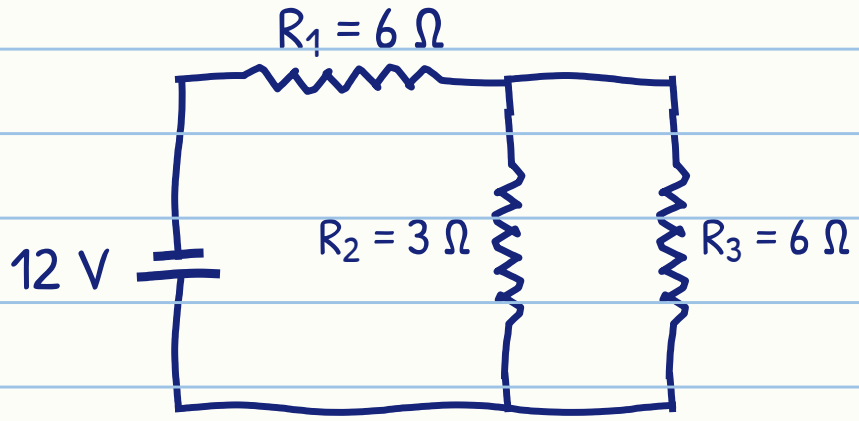
$t \approx 1.57$ s

Horizontal: $x = v_x t = 5(1.57) \approx 7.9$ m

Answer: lands after 1.57 s, 7.9 m from the cliff

Problem P3. Series-parallel resistors

A resistor $R_1 = 6\ \Omega$ is in series with a parallel pair $R_2 = 3\ \Omega$ and $R_3 = 6\ \Omega$, across a 12 V battery. Draw the circuit, then find the equivalent resistance and the current drawn from the battery.



$R_2 \parallel R_3$, then in series with R_1

$R_{23} = R_2 + R_3 = 3 + 6 = 9\ \Omega$

$R_{eq} = R_1 + R_{23} = 6 + 9 = 15\ \Omega$

$I = V/R_{eq}$

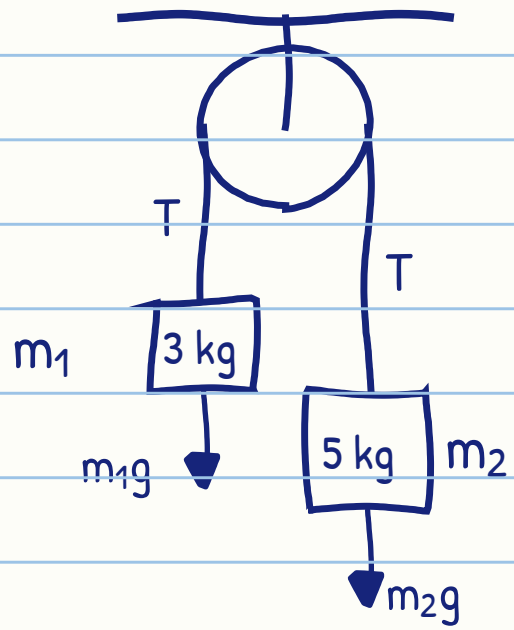
$I = 12/15 = 0.8\ A$

check: $V_1 = IR_1 = 0.8(6) = 4.8\ V$

Answer: $R_{eq} = 15\ \Omega$, $I = 0.8\ A$

Problem P4. Atwood machine

Two masses, $m_1 = 3$ kg and $m_2 = 5$ kg, hang on either side of a light frictionless pulley (an Atwood machine). Find the acceleration of the masses and the tension in the string. Take $g = 9.8$ m/s^2.



$m_2 = 5$ kg goes down, $m_1 = 3$ kg goes up

net force: $(m_2 - m_1)g = 2(9.8) = 19.6$ N

$$a = \frac{(m_2 - m_1)g}{m_2} = \frac{19.6}{5}$$

$a = 3.92$ m/s^2

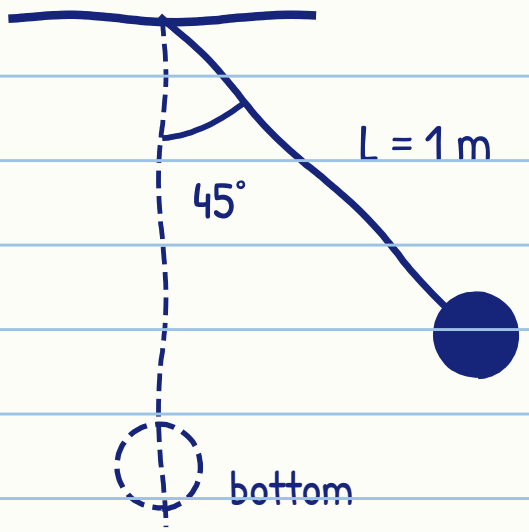
for m_1: $T - m_1 g = m_1 a$

$T = 3(9.8 + 3.92) = 3(13.72) = 41.16$ N

Answer: $a \approx 3.92$ m/s^2, $T \approx 41.2$ N

Problem P5. Pendulum speed at the bottom

A pendulum of length $L = 1$ m is released from rest at 45° from the vertical. Find the speed of the bob at the lowest point. Ignore air resistance; take $g = 9.8$ m/s^2.



released from rest, no friction: energy conserved

$mgh = \frac{1}{2}mv^2$, so $v = \sqrt{2gh}$

height drop $h = L\cos 45^\circ = 1(0.707) = 0.707$ m

$v = \sqrt{2(9.8)(0.707)}$

$= \sqrt{13.86}$

Answer: $v \approx 3.72$ m/s